A.) 3-Stages of Problem Identification

→Domain : Machine Learning

- Problem : Predict Insurance charges

→ Supervised Learning Model : ML based on data

- Input(age,sex,bmi,smoker,etc..)
- Output(charges)

→Regression Model

- Since output is number(continuous value)

B.) To Find following the Machine Learning Regression method using in r2 value

1.Mutiple Linear Regression (R^2 Value) = 0.7894

2.Support Vector Machine

S.NO	Hyper tuning parameter	RBF (R^2 Value)	Poly (R^2 Value)	Sigmoid (R^2 Value)	Linear (R^2 Value)
1	C - 10	-0.0804	-1163	-0.0131	-113.048
2	C - 100	-0.0212	-3297	-0.5435	-146.143
3	C - 1000	-0.0158	-1054	-4.5499	-156.872
4	C - 2000	-0.0490	-4507	-16.3386	-156.696
5	C - 3000	-0.0654	-2700	-58.5396	-156.716

The SVM Regression (R^2 Value) = -0.5435 (Sigmoid,C-100)

3.Decision Tree

S.NO	Criterion	Splitter	R^2 Value
1	Squared_error	best	0.7036
2	Squared_error	random	0.7547
3	Friedman_mse	best	0.6873
4	Friedman_mse	random	0.7246
5	Absolute_error	best	0.6727
6	Absolute_error	random	0.7536
7	Poisson	best	0.7154
8	Poisson	random	0.7075

The Decision Tree (R^2 Value) = 0.7547 (Squared_error,random)

4.Random Forest

S.NO	n - estimators	Squared_ error (R^2 Value)	Friedman_mse (R^2 Value)	Absolute_ error (R^2 Value)	Poisson (R^2 Value)
1	100	0.8516	0.8544	0.8598	0.8547
2	500	0.8565	0.8548	0.8556	0.8558
3	1000	0.8554	0.8555	0.8556	0.8556
4	2000	0.8555	0.8546	0.8551	0.8566
5	3000	0.8557	0.8545	0.8554	0.8556

The Random Forest (R^2 Value) = 0.8598 (absolute_error,n_estimators=100)

C.) The Finale Machine Learning best method of Regression

- ✓ Random Forest (R^2 Value) = 0.8598 (absolute_error,n_estimators=100)